

2016 HSC Mathematics Extension 1 Marking Guidelines

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	B
3	A
4	C
5	A
6	D
7	C
8	B
9	A
10	A

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Obtains $x = y^3 - 2$, or equivalent merit	1

Sample answer:

$$y = x^3 - 2$$

Interchange x and y

$$x = y^3 - 2$$

$$\sqrt[3]{x+2} = y$$

$$\therefore f^{-1}(x) = \sqrt[3]{x+2}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains correct integral in terms of u only, or equivalent merit	2
• Attempts to relate du and dx , or equivalent merit	1

Sample answer:

Let $u = x - 4$

$$\frac{du}{dx} = 1 \Rightarrow du = dx$$

$$\begin{aligned}
 \therefore \int x\sqrt{x-4} \, dx &= \int (u+4)u^{\frac{1}{2}} \, du \\
 &= \int \left(u^{\frac{3}{2}} + 4u^{\frac{1}{2}}\right) \, du \\
 &= \frac{2}{5}u^{\frac{5}{2}} + \frac{8}{3}u^{\frac{3}{2}} + C \\
 &= \frac{2}{5}(x-4)^{\frac{5}{2}} + \frac{8}{3}(x-4)^{\frac{3}{2}} + C \\
 &= \frac{2}{5}\sqrt{(x-4)^5} + \frac{8}{3}\sqrt{(x-4)^3} + C
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Obtains $\frac{k}{a^2 + (bx)^2}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{d}{dx}(3 \tan^{-1}(2x)) &= 3 \cdot \frac{2}{1 + (2x)^2} \\ &= \frac{6}{1 + 4x^2}\end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Observes $2 \sin x \cos x = \sin 2x$, or equivalent merit	1

Sample answer:Method 1

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{2 \sin x \cos x}{3x} \right) &= \frac{2}{3} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \cdot \lim_{x \rightarrow 0} \left(\frac{\cos x}{x} \right) \\ &= \frac{2}{\underline{\underline{3}}}\end{aligned}$$

Method 2

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{2 \sin x \cos x}{3x} \right) &= \frac{2}{3} \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \\ &= \frac{2}{\underline{\underline{3}}}\end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Finds the three boundary values for x , or equivalent merit	2
• Attempts to write as a single fraction, or equivalent merit	1

Sample answer:

$$\frac{3}{2x+5} - x > 0$$

$$\frac{3}{2x+5} > x$$

$$(2x+5)^2 \times \frac{3}{\cancel{(2x+5)}} > x(2x+5)^2 \quad \left(x \neq -\frac{5}{2} \right)$$

$$3(2x+5) > x(2x+5)^2$$

$$x(2x+5)^2 - 3(2x+5) < 0$$

$$(2x+5)(2x^2+5x-3) < 0$$

$$(2x+5)(2x-1)(x+3) < 0$$

$$x < -3 \quad \text{or} \quad -\frac{5}{2} < x < \frac{1}{2}$$

Question 11 (e) (continued)

Method 2

$$\frac{3}{2x+5} - x > 0$$

Solve $\frac{3}{2x+5} - x = 0$ for $x \neq -\frac{5}{2}$

$$3 - x(2x+5) = 0$$

$$2x^2 + 5x - 3 = 0$$

$$(2x-1)(x+3) = 0$$

$$x = \frac{1}{2}, \quad x = -3$$



Test	$x = -4$	$x = 0$	$x = 1$
	$\frac{3}{-3} + 4 > 0$	$\frac{3}{5} > 0$	$\frac{3}{7} - 1 < 0$
	✓	✓	✗

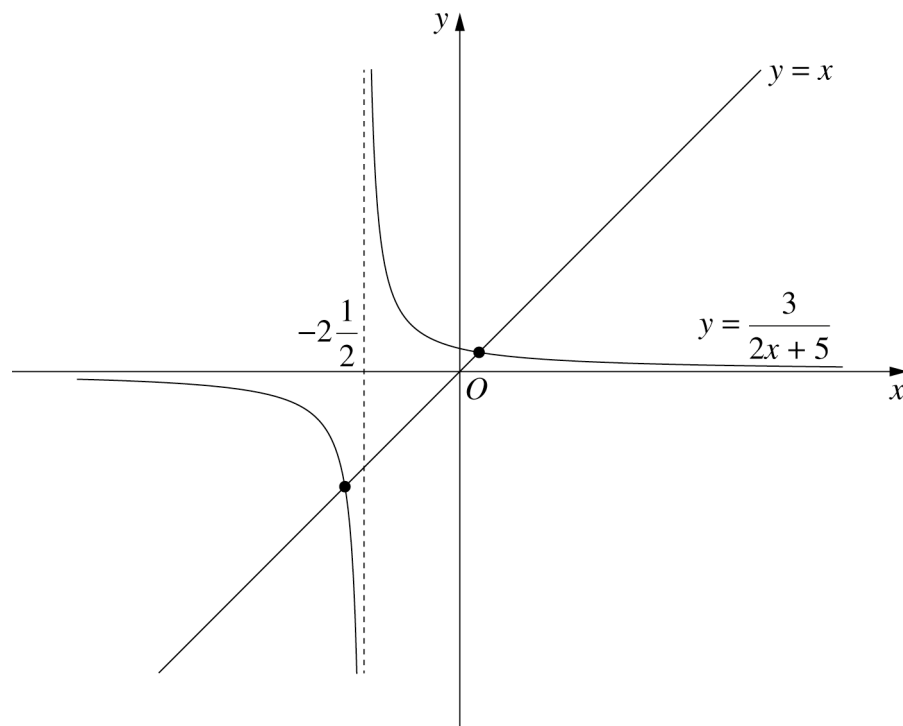
$$x < -3 \text{ or } -\frac{5}{2} < x < \frac{1}{2}$$

Question 11 (e) (continued)

Method 3

$$\frac{3}{2x+5} - x > 0$$

$$\frac{3}{2x+5} > x$$



$$\begin{aligned} 3 &= 2x^2 + 5x \\ 2x^2 + 5x - 3 &= 0 \\ (2x - 1)(x + 3) &= 0 \\ x &= \frac{1}{2}, -3 \end{aligned}$$

$$\text{Solution: } x < -3 \text{ or } -2\frac{1}{2} < x < \frac{1}{2}$$

Question 11 (f) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$P(S) = \frac{3}{5}$$

The possibilities are SNN, NNS, NSN

$$\begin{aligned}\therefore P(1S + 2N) &= \left(\frac{3}{5} \times \frac{2}{5} \times \frac{2}{5}\right) \times 3 \\ &= \frac{36}{125}\end{aligned}$$

Question 11 (f) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use complementary events, or equivalent merit	1

Sample answer:Method 1

$$\begin{aligned}P(\text{at least 2 bullseyes}) &= {}^6C_2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^4 + {}^6C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^3 + {}^6C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^2 \\ &\quad + {}^6C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^1 + {}^6C_6 \left(\frac{3}{5}\right)^6 \left(\frac{2}{5}\right)^0 \\ &= \frac{432}{3125} + \frac{864}{3125} + \frac{972}{3125} + \frac{2916}{15625} + \frac{729}{15625} \\ &= \frac{2997}{3125}\end{aligned}$$

Method 2

$$\begin{aligned}P(\text{at least 2 bullseyes}) &= 1 - \left[\left(\frac{2}{5}\right)^6 + {}^6C_1 \left(\frac{3}{5}\right) \left(\frac{2}{5}\right)^5 \right] \\ &= \frac{2997}{3125}\end{aligned}$$

Question 12 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

By similar triangles,

$$\frac{r}{5} = \frac{h}{20}$$

$$\therefore r = \frac{h}{\underline{\underline{4}}}$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$v = \frac{1}{3}\pi r^2 h$$

$$v = \frac{1}{3}\pi \left(\frac{h}{4}\right)^2 h$$

$$v = \frac{\pi}{48} h^3$$

$$\therefore \frac{dv}{dh} = \frac{3\pi h^2}{48}$$

$$\Rightarrow \frac{dv}{dh} = \frac{\pi}{\underline{\underline{16}}} h^2$$

Question 12 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use $\frac{dA}{dt}$, or equivalent merit	1

Sample answer:

Show $\frac{dh}{dt} = \frac{-0.32}{\pi h}$

$$A = \pi r^2$$

$$r = \frac{h}{4}$$

$$A = \frac{\pi h^2}{16}$$

$$\frac{dA}{dh} = \frac{\pi h}{8}$$

$$\frac{dA}{dt} = \frac{\pi h}{8} \times \frac{dh}{dt} = -0.04$$

$$\therefore \frac{dh}{dt} = \frac{-0.32}{\pi h}$$

Question 12 (a) (iv)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the chain rule, or equivalent merit	1

Sample answer:

$$\frac{dv}{dh} = \frac{\pi}{16} h^2$$

$$\frac{dh}{dt} = \frac{-0.32}{\pi h}$$

$$\frac{dv}{dt} = \frac{dv}{dh} \cdot \frac{dh}{dt}$$

$$= \frac{\cancel{\pi} h^2}{16} \cdot \frac{-0.32}{\cancel{\pi} h}$$

$$= -0.02h$$

When $h = 10$, $\frac{dv}{dt} = -0.2$

$\therefore$ the volume decreases at a rate of $0.2 \text{ cm}^3 \text{ s}^{-1}$

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Obtains $\frac{dx}{dt} = k(500 - x)$, or equivalent merit	2
• Obtains $\frac{dx}{dt} = ky$ or $y = 500 - x$, or equivalent merit	1

Sample answer:

$$x + y = 500$$

$$y = 500 - x$$

$$\frac{dx}{dt} \propto y$$

$$\therefore \frac{dx}{dt} = ky \quad \text{where } k \text{ is a constant}$$

$$\Rightarrow \frac{dx}{dt} = k(500 - x)$$

When $x = 0$, $\frac{dx}{dt} = 2$

$$\therefore 2 = k(500 - 0)$$

$$\Rightarrow k = \underline{\underline{0.004}}$$

$$\therefore \frac{dx}{dt} = \underline{\underline{0.004(500 - x)}}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Differentiates the given expression, or equivalent merit	1

Sample answer:Method 1

$$x = 500 - Ae^{-0.004t}$$

$$\frac{dx}{dt} = A \times 0.004e^{-0.004t}$$

$$= 0.004(Ae^{-0.004t})$$

$$= 0.004(500 - x) \quad \text{as required.}$$

Now, $t = 0$, $\frac{dx}{dt} = 2$, $x = 0$

$$\therefore 0 = 500 - Ae^{-0.004(0)}$$

$$A = 500$$

Method 2

$$\frac{dx}{dt} = 0.004(500 - x)$$

$$\int \frac{dx}{500 - x} = \int 0.004 dt$$

$$-\ln(500 - x) = 0.004t + C$$

$$500 - x = Ae^{-0.004t} \quad \text{where } A \text{ is a constant.}$$

$$\therefore x = \underline{\underline{500 - Ae^{-0.004t}}}$$

When $t = 0$, $x = 0$, $\Rightarrow A = \underline{\underline{500}}$

$$\therefore x = \underline{\underline{500(1 - e^{-0.004t})}}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Relates the slopes of both curves at α , or equivalent merit	1

Sample answer:

$$y = \tan x$$

$$\therefore \frac{dy}{dx} = \sec^2 x$$

$$\text{At } x = \alpha, \quad \frac{dy}{dx} = \sec^2 \alpha$$

$$y = \cos x$$

$$\therefore \frac{dy}{dx} = -\sin x$$

$$\text{At } x = \alpha, \quad \frac{dy}{dx} = -\sin \alpha$$

$$\text{Now product of gradients} = \sec^2 \alpha \times (-\sin \alpha)$$

$$= -\frac{\sin \alpha}{\cos \alpha} \cdot \frac{1}{\cos \alpha}$$

$$= -\frac{\tan \alpha}{\cos \alpha}$$

$$= -1 \quad \text{since } \tan \alpha = \cos \alpha \text{ at the point of intersection}$$

$\therefore$ Tangents to the curves are perpendicular at $x = \alpha$.

Question 12 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds a relevant function to use Newton's method, or equivalent merit	1

Sample answer:Let $f(x) = \tan x - \cos x$

$$f'(x) = \sec^2 x + \sin x$$

Using Newton's method to solve $\tan \alpha = \cos \alpha$, with $x_0 = 1$.

$$\begin{aligned}
 x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= 1 - \frac{f(1)}{f'(1)} \\
 &= 1 - \frac{\tan 1 - \cos 1}{\sec^2 1 + \sin 1} \\
 &\approx 1 - \frac{1.0171054}{4.2669898} \\
 &\approx 1 - 0.238366 \\
 &= \underline{0.76} \quad (2 \text{ decimal places})
 \end{aligned}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct explanation	2
• Explains why the amplitude is 4, or equivalent merit	1

Sample answer:

From reference sheet

$$x = b + a \cos(nt + \alpha)$$

High tide = maximum = 9

Low tide = minimum = 1

$$\therefore a = \frac{9-1}{2} = 4$$

$$\text{Period, } T = \frac{2\pi}{n}; \quad \therefore \frac{25}{2} = \frac{2\pi}{n}$$

$$\Rightarrow n = \frac{4\pi}{25}$$

$t = 0$ at the high tide so $b = 5$ and $\alpha = 0$

$$\therefore x = 5 + 4 \cos\left(\frac{4\pi t}{25}\right)$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to find t when $\ddot{x} = 0$, or equivalent merit	1

Sample answer:

$$x = 5 + 4 \cos\left(\frac{4\pi t}{25}\right)$$

$$\frac{dx}{dt} = -4 \cdot \frac{4\pi}{25} \sin\left(\frac{4\pi t}{25}\right)$$

$$\frac{d^2x}{dt^2} = \frac{-16\pi}{25} \cdot \frac{4\pi}{25} \cos\left(\frac{4\pi t}{25}\right)$$

Fastest when $\frac{d^2x}{dt^2} = 0$

Increasing fastest when

$$\cos\left(\frac{4\pi t}{25}\right) = 0$$

$$\therefore \frac{4\pi t}{25} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$t = \frac{25}{8}, \frac{75}{8}, \dots$$

Tide is going *out* when $t = \frac{25}{8}$

$$\therefore \text{Increasing when } t = \underline{\underline{\frac{75}{8}}} = 9\frac{3}{8}.$$

So $9\frac{3}{8}$ hours after 2 am makes the time $11:22\frac{1}{2}$ am (11:22:30 am)(or $11\frac{3}{8}$ hours after midnight).

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Finds $\dot{y}$, or equivalent merit	1

Sample answer:

$$x = ut \cos \theta \quad y = ut \sin \theta - 5t^2$$

$$\dot{x} = u \cos \theta \quad \dot{y} = u \sin \theta - 10t$$

Maximum height if $\dot{y} = 0 \Rightarrow u \sin \theta - 10t = 0$

$$u \sin \theta = 10t$$

$$t = \frac{u \sin \theta}{10}$$

Substitute into $y = ut \sin \theta - 5t^2$

$$\begin{aligned} \therefore y &= u \sin \theta \cdot \frac{u \sin \theta}{10} - 5 \left(\frac{u \sin \theta}{10} \right)^2 \\ &= \frac{u^2 \sin^2 \theta}{10} - \frac{u^2 \sin^2 \theta}{20} \end{aligned}$$

$$\therefore \text{Maximum height} = \frac{u^2 \sin^2 \theta}{20}$$

Question 13 (b) (ii)

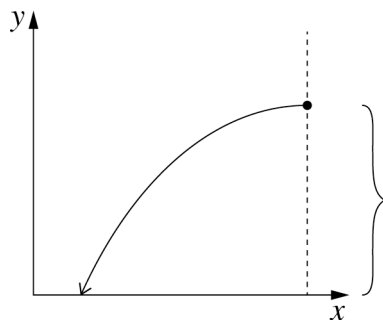
Criteria	Marks
• Provides correct solution	2
• Shows that the maximum height above the projection point is $\frac{45}{4}$, or equivalent merit	1

Sample answer:

$$\begin{aligned} \text{Maximum height} &= 20 + \frac{30^2 \sin^2 30}{20} \\ &= 20 + \frac{900 \times \frac{1}{4}}{20} \\ &= \frac{125}{4} \end{aligned}$$

Question 13 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains correct equation for y in terms of t , or equivalent merit	1

Sample answer:

Taking left ($\leftarrow$) to be positive and origin as start point.

$$\dot{x} = 10 \quad \ddot{y} = -10$$

$$x = 10t \quad \dot{y} = -10t \quad (\text{since } \dot{y}(0) = 0)$$

$$(\text{since } x(0) = 0) \quad y = -5t^2 \quad (\text{since } y(0) = 0)$$

$$y = -31.25 \Rightarrow -31.25 = -5t^2$$

$$t^2 = 6.25$$

$$t = \pm 2.5, \quad t > 0$$

$$t = 2.5$$

$\therefore$ Ball hits the ground after 2.5 seconds.

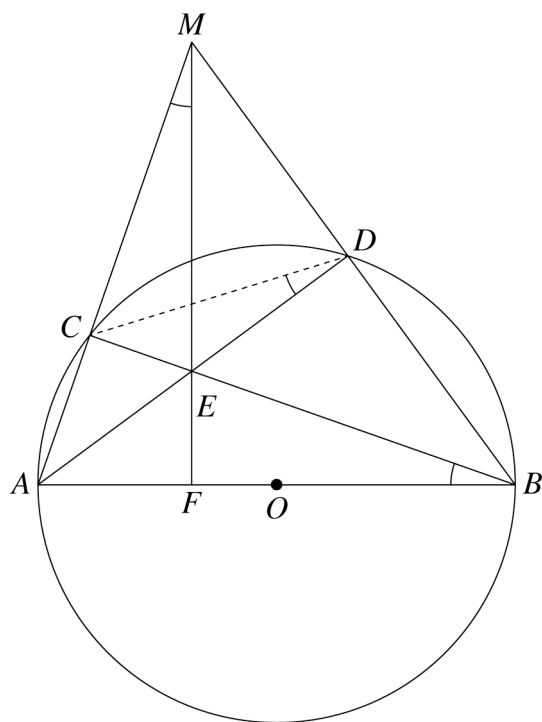
Question 13 (b) (iv)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$t = 2.5 \Rightarrow x = 10(2.5) \\ = 25$$

$\therefore$ The ball lands 25 m from the base of the wall.

Question 13 (c)**Question 13 (c) (i)**

Criteria	Marks
• Provides correct solution	2
• Shows that one of the angles at C or D is a right angle, or equivalent merit	1

Sample answer:

$$\angle ADB = 90^\circ, \quad \angle BCA = 90^\circ \quad (\angle \text{ in a semicircle})$$

$$\angle MDE = 90^\circ, \quad (\angle \text{ sum, straight line})$$

$\therefore CMDE$ is a cyclic quadrilateral (Exterior $\angle ACB$ equal to opposite interior $\angle MDE$)

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows $\angle ABC = \angle ADC$, or equivalent merit	1

Sample answer:Method 1Construct CD Let $\angle ABC = x^\circ$

$$\therefore \angle ADC = x^\circ \quad (\angle\text{s in the same segment})$$

Similarly $\angle CME = x^\circ$

$$\angle CAB = 90^\circ - x^\circ \quad (\angle \text{sum } \triangle ABC)$$

$$\begin{aligned} \therefore \angle MFA &= 180^\circ - \angle CME - \angle CAB && (\angle \text{sum } \triangle MFA) \\ &= 180^\circ - (x^\circ) - (90^\circ - x^\circ) \\ &= 90^\circ \end{aligned}$$

$$\therefore MF \perp AB$$

Method 2Construct CD as shown in the diagram

$$\angle ABC = \angle ADC \quad (\angle\text{s in the same segment})$$

$$\angle ADC = \angle EMC \quad (\angle\text{s in the same segment as } CDME \text{ are concyclic points})$$

$$\triangle CEM \parallel \triangle FEB \quad (\text{equiangular})$$

$$\therefore \angle MCE = \angle BFE = 90^\circ \quad (\text{corresponding } \angle\text{s})$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}\text{R.T.P. } 4n^3 + 18n^2 + 23n + 9 \\ = (n+1)(4n^2 + 14n + 9)\end{aligned}$$

$$\begin{aligned}\text{RHS} &= n(4n^2 + 14n + 9) + 1(4n^2 + 14n + 9) \\ &= 4n^3 + 14n^2 + 9n + 4n^2 + 14n + 9 \\ &= 4n^3 + 18n^2 + 23n + 9 \\ &= \text{LHS}\end{aligned}$$

OR

$$\begin{aligned}\text{LHS} &= 4n^3 + 4n^2 + 14n^2 + 14n + 9n + 9 \\ &= 4n^2(n+1) + 14n(n+1) + 9(n+1) \\ &= (n+1)(4n^2 + 14n + 9) \\ &= \text{RHS}\end{aligned}$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct proof	3
• Establishes induction step, or equivalent merit	2
• Establishes initial case, or equivalent merit	1

Sample answer:

$$\text{Let } T_n = (2n-1)(2n+1), S_n = \frac{1}{3}n(4n^2 + 6n - 1)$$

$$\text{R.T.P. } 1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1)$$

$$\begin{aligned} \text{When } n = 1, \quad \text{LHS} &= 1 \times 3 & \text{RHS} &= \frac{1}{3}(1)(4(1)^2 + 6(1) - 1) \\ &= 3 & &= \frac{1}{3} \times 9 \\ & & &= 3 \end{aligned}$$

 $\therefore$ True for $n = 1$

$$\text{Assume true for some integer } k, \text{ ie } S_k = \frac{1}{3}k(4k^2 + 6k - 1)$$

and attempt to show it is also true for $k + 1$

$$\begin{aligned} \text{ie, try to show } S_{k+1} &= \frac{1}{3}(k+1)(4(k+1)^2 + 6(k+1) - 1) \\ &= \frac{1}{3}(k+1)(4k^2 + 8k + 4 + 6k + 6 - 1) \\ &= \frac{1}{3}(k+1)(4k^2 + 14k + 9) \end{aligned}$$

$$\text{Now } S_{k+1} = S_k + T_{k+1}$$

$$\begin{aligned} &= \frac{1}{3}k(4k^2 + 6k - 1) + (2(k+1)-1)(2(k+1)+1) \\ &= \frac{1}{3}k(4k^2 + 6k - 1) + (2k+1)(2k+3) \\ &= \frac{1}{3}(4k^3 + 6k^2 - k) + 4k^2 + 8k + 3 \\ &= \frac{1}{3}(4k^3 + 6k^2 - k + 12k^2 + 24k + 9) \\ &= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9) \\ &= \frac{1}{3}(k+1)(4k^2 + 14k + 9) \quad (\text{From part (i)}) \end{aligned}$$

as required

 $\therefore$ Hence the result is true for all positive integers n , by induction.

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$$

Let $x = 1$ then

$$(1+1)^n = \binom{n}{0} + \binom{n}{1}(1) + \binom{n}{2}(1)^2 + \cdots + \binom{n}{n}(1)^n$$

$$\therefore 2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:Differentiate $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$, with respect to x .

$$n(1+x)^{n-1} = \binom{n}{1} + \binom{n}{2}(2x) + \binom{n}{3}(3x^2) + \cdots + \binom{n}{n}nx^{n-1}$$

and substitute $n = 1$ into this equation

$$n(1+1)^{n-1} = \binom{n}{1} + \binom{n}{2}(2(1)) + \binom{n}{3}(3(1)^2) + \cdots + \binom{n}{n}(n(1)^{n-1})$$

$$\therefore n \cdot 2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Attempts to write the right-hand side in terms of the previous two parts, or equivalent merit	1

Sample answer:Method 1

$$\begin{aligned}
\sum_{r=1}^n \binom{n}{r} (2r - n) &= \binom{n}{1}(2 - n) + \binom{n}{2}(4 - n) + \binom{n}{3}(6 - n) + \cdots + \binom{n}{n}(2n - n) \\
&= 2\binom{n}{1} + 4\binom{n}{2} + 6\binom{n}{3} + \cdots + 2n\binom{n}{n} - n\left[\binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}\right] \\
&= 2\left\{\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + n\binom{n}{n} - n\left[\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}\right] - \binom{n}{0}\right\} \\
&= 2(n \cdot 2^{n-1}) - n(2^n - 1) \\
&= n2^n - n2^n + n \\
&= n
\end{aligned}$$

Method 2

$$\begin{aligned}
&\sum_{r=1}^n \binom{n}{r} (2r - n) \\
&= 2\sum_{r=1}^n \binom{n}{r} r - n\sum_{r=1}^n \binom{n}{r} \\
&= 2\sum_{r=0}^n \binom{n}{r} r - n\left(\sum_{r=0}^n \binom{n}{r} - 1\right) \\
&= 2 \cdot n \cdot 2^{n-1} - n(2^n - 1) \\
&= n
\end{aligned}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

From reference sheet, equation of tangent is $y = tx - at^2$

At D on the directrix, $y = -a$

$$\Rightarrow -a = tx - at^2$$

$$tx = at^2 - a$$

$$x = at - \frac{a}{t}$$

$$\therefore D \text{ is } \left(at - \frac{a}{t}, -a \right)$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Attempts to eliminate the parameter to find the Cartesian equation of $\mathcal{P}_2$, or equivalent merit	2
• Finds coordinates of R , or equivalent merit	1

Sample answer:

From reference sheet, equation of normal is $x + ty = at^3 + 2at$

At R (above D), $x = at - \frac{a}{t}$

$$\Rightarrow at - \frac{a}{t} + ty = at^3 + 2at$$

$$ty = at^3 + at + \frac{a}{t}$$

$$y = at^2 + a + \frac{a}{t^2}$$

$$= a\left(t^2 + 1 + \frac{1}{t^2}\right)$$

$$\therefore R \text{ is } \left(a\left(t - \frac{1}{t}\right), a\left(t^2 + 1 + \frac{1}{t^2}\right)\right)$$

Method 1

$$x = a\left(t - \frac{1}{t}\right)$$

$$x^2 = a^2\left(t^2 - 2 + \frac{1}{t^2}\right)$$

$$\frac{x^2}{a^2} = t^2 + \frac{1}{t^2} - 2$$

$$\frac{x^2}{a^2} + 2 = t^2 + \frac{1}{t^2}$$

$$y = a\left(\left(t^2 + \frac{1}{t^2}\right) + 1\right)$$

$$= a\left(\frac{x^2}{a^2} + 2 + 1\right)$$

$$y = \frac{x^2}{a} + 3a$$

$$ay = x^2 + 3a^2$$

Method 2

$$\frac{x}{a} = t - \frac{1}{t}$$

$$y = a\left(t^2 + 1 + \frac{1}{t^2}\right)$$

$$= a\left(\left(t - \frac{1}{t}\right)^2 + 3\right)$$

$$= a\left(\frac{x^2}{a^2} + 3\right) = \frac{x^2}{a} + 3a$$

$$\therefore (y - 3a)a = x^2$$

Locus is $x^2 = a(y - 3a)$.

A parabola! Vertex $(0, 3a)$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

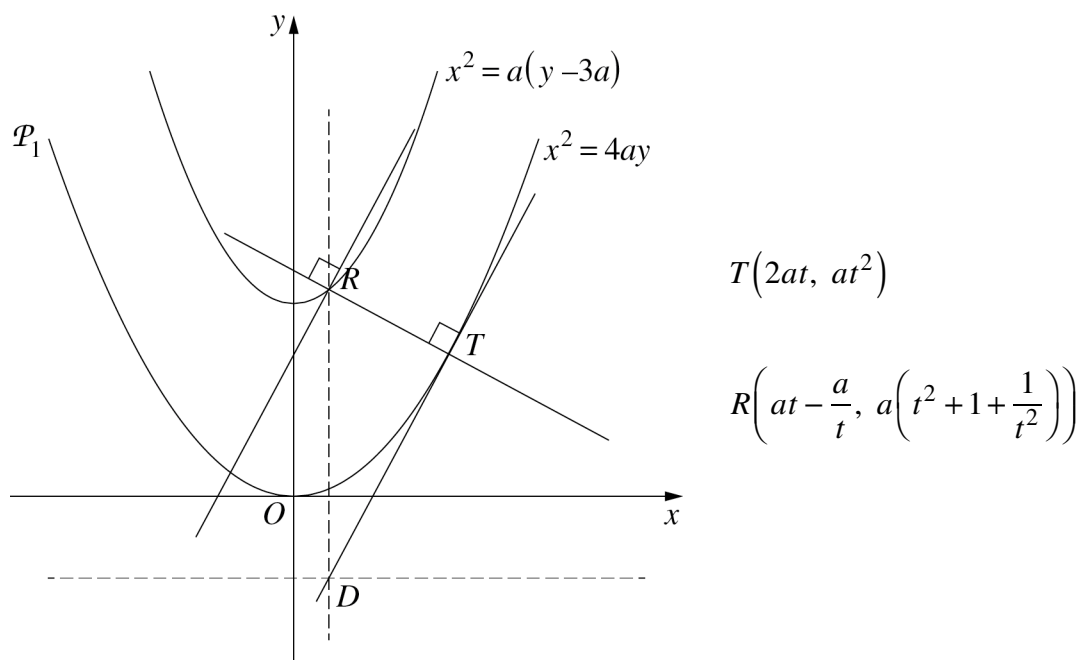
$$x^2 = a(y - 3a)$$

$$\text{ie } x^2 = 4\left(\frac{a}{4}\right)(y - 3a)$$

$$\therefore \text{Focal length } \frac{a}{4}$$

Question 14 (c) (iv)

Criteria	Marks
• Provides correct solution	2
• Finds slope of $\mathcal{P}_2$ at R , or equivalent merit	1

Sample answer:

A common normal will occur when the tangents at T and R are parallel.

Now gradient at T is t

$$\text{For } R, \quad \frac{x^2}{a} = y - 3a$$

Question 14 (c) (iv) continued

$$y = \frac{x^2}{a} + 3a$$

$$y' = \frac{2x}{a}$$

$$\text{So } y' = \frac{2\left(at - \frac{a}{t}\right)}{a}, \text{ since } x = at - \frac{a}{t},$$

$$= 2\left(t - \frac{1}{t}\right)$$

$$\text{So gradient at } R \text{ is } 2t - \frac{2}{t}$$

$$\text{For tangents to be parallel, } t = 2t - \frac{2}{t}$$

$$t^2 = 2t^2 - 2$$

$$2 = t^2$$

$$t = \pm\sqrt{2}$$

2016 HSC Mathematics Extension 1

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	7.1	H5, HE7
2	1	16.2E	PE3
3	1	5.7E	P3, PE6
4	1	2.10E, 2.9E	PE3
5	1	13.6E	HE6
6	1	15.2E, 5.9E	HE4
7	1	14.3, 14.4E	HE5
8	1	18.1E	PE3
9	1	10.1, 10.4	H6
10	1	16.1 E, 16.3E	PE3

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	15.1E	HE4
11 (b)	3	11.5E	HE6
11 (c)	2	15.5E	HE4
11 (d)	2	5.7E, 13.4E	HE7
11 (e)	3	1.4E	PE3
11 (f) (i)	1	3.2, 18.2E	HE3
11 (f) (ii)	2	18.2E	HE3
12 (a) (i)	1	2.3	P2, PE6
12 (a) (ii)	1	14.1E	HE5
12 (a) (iii)	2	14.1E	HE5
12 (a) (iv)	2	14.1E	HE5
12 (b) (i)	3	14.2E	HE3
12 (b) (ii)	2	14.2E	HE3
12 (c) (i)	2	6.2, 10.7, 13.5	H5, HE4
12 (c) (ii)	2	16.4E	HE7
13 (a) (i)	2	14.4E	HE3
13 (a) (ii)	2	14.4E	HE3

Question	Marks	Content	Syllabus outcomes
13 (b) (i)	2	14.3E	HE3
13 (b) (ii)	2	14.3E	HE3
13 (b) (iii)	2	14.3E	HE3
13 (b) (iv)	1	14.3E	HE3
13 (c) (i)	2	2.10E, 2.7E	PE3
13 (c) (ii)	2	2.10E, 2.8E	PE3
14 (a) (i)	1	1.3	P4, PE6
14 (a) (ii)	3	7.4E	HE2
14 (b) (i)	1	17.1E, 17.3E	HE7
14 (b) (ii)	1	8.7, 17.1E, 17.3E	HE4, HE7
14 (b) (iii)	2	17.1E, 17.3E	HE7
14 (c) (i)	1	9.6E	PE3
14 (c) (ii)	3	9.6E	PE3
14 (c) (iii)	1	9.6E	PE3
14 (c) (iv)	2	10.7, 9.6E	HE7